BrightLearn Data Analytics

Research Assignment 1

Foundations of Data Analytics & Data Science

Course	Data Analytics & Data Science
Assignment No.	Research Assignment 1
Target Audience	Beginner Students
Due Date	To be confirmed by instructor
Format	Handwritten answers — scan and email your completed work

Instructions

Read each question carefully and research the answer before writing.

- All answers must be handwritten — typed submissions will not be accepted.
- Write in full sentences and use your own words.
- Include diagrams or examples where relevant — this will earn you bonus marks.
- Cite your sources at the bottom of each section.
- Scan your completed assignment and submit it before the due date.

Section A: Data Fundamentals

Research Tip: *Start with official documentation from platforms like IBM, Google Cloud, AWS, or Microsoft Azure — they offer excellent beginner-friendly explanations.*

1. What is a Database? Define it in your own words and explain why databases are essential in the modern world. Provide two real-world examples of where databases are used.

A Database is an organised collection of data that is stored electronically so it can be easily accessed, managed & updated. They are essential in the modern world because

almost every single digital System depends on them to function efficiently. They help business Store Customer information, track transactions, and make decisions based

Real world examples : - A banking system uses a database to Store account balances and trasactions.

- A Social media platform stores user profiles, posts and messages in a database.

2. What is a Data Warehouse? How does it differ from a regular database? Why would a company use a data warehouse?

A Datawarehouse is a system used to store large amounts of historical data from different sources for analysis and reporting.

Regular database :- handles daily operations
Data warehouse :- Used for analysis of historical data.

Companies use data warehouses to analyse trends, Create reports and support business decisions.

3. What is a Data Lake? Explain the concept and describe how it differs from a data warehouse. When would a business prefer a data lake over a data warehouse?

A Data Lake is a storage system that holds large amounts of raw data in it's original format including structured and unstructured data.

Differences :- Data Warehouse → Structured, cleaned data
Data Lake → raw, unprocessed data.

Businesses use data lakes when they want to store all data without processing it first. Also if they work with big data.

4. What is a Data Mart? How does it relate to a data warehouse and who typically uses it?

A Data Mart is a smaller, focused version of data warehouse that is designed for a specific department or business area.

It is usually used by Sales teams, Marketing teams or finance departments.

5. Explain the difference between Structured, Semi-Structured, and Unstructured Data. Provide one example of each.

Structured Data :- Data that is organised in tables with rows and columns
Example : Customer database.

Semi-Structured Data :- Data that is partially organised but not in strict tables.
Example : JSON files.

Unstructured Data :- Data that has no clear format.
Example : Videos, Images, audio files

6. What is Data Modeling? Why is it important when designing a system that works with data? Describe at least two types of data models.

Data Modelling is the process of designing how data will be structured, stored, and related in a database.
it is important because it ensures data is organised properly, systems run efficiently and relationships between data are clear.
Types of data models: Conceptual model → high-level design.
Logical model → defines structure and relationships.

7. What is Data Mining? How is it used in business? Provide two real-world examples of data mining in action.

Data Mining is the process of analysing large datasets to discover patterns, trends, and useful information.

Examples:
- Retail stores analysing buying patterns
- Banks detecting fraud.

8. What is a Database Management System (DBMS)? Name and briefly describe four popular DBMS platforms (e.g., MySQL, PostgreSQL, Oracle, MongoDB) and explain a suitable use case for each.

A Database Management System is a software used to create, manage, and control access to databases.

Examples:
- MySQL → used for web applications
- PostgreSQL → advanced data analysis
- Oracle → large enterprise systems.
- MongoDB → handles unstructured data.

9. What is ETL (Extract, Transform, Load)? Explain each step and describe why this process is important in data analytics.

ETL stands for:
- Extract → collect data
- Transform → Clean and organise data.
- Load → Store data in a system.

It is important because it ensures data is accurate and ready for analysis.

10. What are common data storage formats used in analytics (e.g., CSV, JSON, Parquet, Avro)? Briefly describe each and state when you would use one over another.

- CSV → Simple table format
- JSON → flexible, used in APIs
- Parquet → efficient for big data.
- Avro → used in data Streaming.

Section B: Artificial Intelligence & Machine Learning

Research Tip: *Explore resources like Google's Machine Learning Crash Course, IBM's AI explanations, or OpenAI's blog to build a strong foundational understanding.*

11. What is Artificial Intelligence (AI)? Define it and explain the difference between Narrow AI (Weak AI) and General AI (Strong AI). Provide one example of each.

AI refers to the ability of machines or computer systems to perform tasks that norma require human intelligence.

Types of AI: Narrow AI → Designed to perform a specific task only.
E.g. Voice assistants like SIRI or Google Assistant.

General AI → This is an advanced AI that perform intellectual tasks a human can do. E.g. Fully human-like robots.

12. What is Machine Learning (ML)? How does it differ from traditional programming? Describe the three main types of Machine Learning: Supervised, Unsupervised, and Reinforcement Learning.

Machine Learning is a type of AI that allows computers to learn from data without being explicitly programmed. The system learns patterns from data and improves overti

Traditional programming → rules are written by humans. Machine Learning → System learn rules from data.

Supervised learning → The model is trained using labled data like email spam detectio

Unsupervised learning → The model finds patterns in data without labels like customer segmentation.

Reinforcemet learning → Model learns through rewards and penalties like robot or gar

13. What is Deep Learning? How does it relate to Machine Learning and Artificial Intelligence? What are Neural Networks and how do they work?

- Deep Learning is a subset of Machine Learning that uses complex algorithms called neural networks to learn from large amounts of data.
- Neural networks are systems inspired by human brain. They consist of layers o connected nodes (neurons) that process information.

Input layer → receives data Hidden layer → process data Output layer → gives results.

14. What is Natural Language Processing (NLP)? Describe what it is and provide three real-world applications of NLP that you use in everyday life.

NLP is a field of AI that enables computers to understand, interpret, and respond t human language.

Real World Applications: ChatGPT, Google Translate and Voice assistant.

15. What is a Large Language Model (LLM)? How is it different from a standard Machine Learning model? Name two well-known LLMs and explain what they are used for.

A LLM is an advanced AI Model trained on massive amounts of text data to understand and generate human-like language. Standard ML focuses on specific tasks and LLM understands and generates language across many task.

Examples: ChatGPT
: Claude

16. What is Generative AI? How does it work at a high level? Name three tools that use Generative AI and describe what each one does.

- Generative AI is a type of AI that can create new content, Such as text, images, music or videos.
- it learns patterns from existing data and uses those patterns to generate new content.
- ChatGPT → generates text; DALL-E → Creates images Midjourney → generate artwork

17. What is a Training Dataset? Why is the quality of training data so important when building an AI model? What could go wrong if the training data is biased or incorrect?

- A Training dataset is the data used to teach an AI or Machine learning model.
- Good data → accurate results Bad data → poor or biased results.
- Bias → unfair decisions incorrect data → Wrong predictions.

18. What is Computer Vision? How do machines 'see' and interpret images? Provide two real-world use cases of computer vision.

- Computer Vision is a field of AI that enables machines to see and interpret images or videos.
- The system analyses pixels in images. it detects patterns, shapes and objects
- Real World use case: Facial recognition and self-driving Cars.

Section C: APIs, Tools & Developer Concepts

Research Tip: *Search for 'What is an API for beginners' on YouTube or read documentation from platforms like Postman, RapidAPI, or Anthropic.*

19. What is an API (Application Programming Interface)? Explain what it is using a simple real-world analogy. Why are APIs important in modern software and data systems?

- An API is a set of rules that allows different software systems to communicate with each other.
- An API is like a waiter in a restaurant. You (user) place and order. The waiteress (API) takes it to the kitchen.
- APIs are important as they allow systems to connect and share data. They enable apps to use external services

20. What is an API Key? What is its purpose? How should it be stored and protected, and what could happen if it is exposed publicly?

- An API key is a unique code used to identify and authenticate a user or application when accessing an API.
- Purpose: Controls access to the API; Tracks usage; Prevents abuse.
- Stored: Keep it private and secure; Store in environment variables (not in code)
- Exposed: Unauthorized users, security risks.

21. What is a REST API? What does REST stand for and what are the main HTTP methods (GET, POST, PUT, DELETE) used in REST APIs? Briefly describe each method.

- Rest API is a type of API that follows specific rules for communication over the internet
- Get: fetch users, POST: create user; PUT: update user; DELETE: delete user

22. What is JSON (JavaScript Object Notation)? Why is it commonly used when working with APIs and data exchange? Write a simple example of a JSON object.

- JSon is a lightweight format used to store and exchange data.
- Used as it is easy to read and write; works well with APIs;
- Lightweight and fast.

```
{
  "name" : "Katlego",
  "age" : 25
  "City" : "Johanessburg"
}
```

23. What is Prompt Engineering? Why is it an important skill when working with AI tools like ChatGPT or Claude? Provide two examples of a poorly written prompt and an improved version of each.

- Prompt engineering is the skill of writing clear and effective instructions to get the best results from AI tools.
- Importance: Better prompt = best results, helps control AI output.
- Poor prompt: "Tell me ~~do~~ about data" Improved: "Explain what a database is in simple terms with examples.
- Poor prompt: "Write something above AI" improved: "Write a 100-word explanation of AI for beginners with two real-world examples.

24. What is a MAP function? Explain what it does in programming and data processing. Provide a simple example in either Python or JavaScript.

A MAP function is used to apply a function to every item in a list and return a new ~~stea~~ list.

25. What is Cloud Computing? Name the three main cloud service models (IaaS, PaaS, SaaS) and briefly explain each. Give one example of a product for each model.

- Cloud Computing is the delivery of computing services on the internet.
- Service Models: Iaas provides virtual ~~machine~~ machines and storage.
  Paas provides a platform for developers to build applications.
  Saas: provides ready-to-use software.

26. What is a Version Control System? What is Git and why is it widely used by data analysts and developers? Briefly explain what a 'commit' and a 'branch' are.

- A VSC is a tool that tracks changes in code over time.
- Git is a popular version control system used by developers and data analysts.
- Used to track changes, allows collaboration and prevents loss of work.
- Commit → Save changes Branch → creating a separate version to work on.

27. What is a Jupyter Notebook? What programming language is it most commonly used with? Why is it popular among Data Analysts and Data Scientists?

- A Jupyter Notebook is an interactive tool used to write and run code, display results and add explanations.
- Python is the commonly used language, it is popular because it combines code and text; easy for data analysis, good for visualisation.

28. What is Python? Why is it considered one of the most popular programming languages for Data Analytics and AI? Name four Python libraries commonly used in data work and briefly describe each.

- Python is a programming language that is widely used for data analysis, AI and automation.
- It is popular because it is easy to learn, powerful, large community.
- Common libraries: Pandas → data analysis
  NumPy → numerical operations.
  Matplotlib → data visualisation.
  Scikit-learn → machine learning.

Section D: Data Ethics, Governance & Careers

Research Tip: *Explore the World Economic Forum, GDPR documentation, and LinkedIn Learning for content on data ethics and career pathways in data.*

29. What is Data Privacy? Why is it important? Research the GDPR (General Data Protection Regulation) and explain what it is and which countries it applies to.

- Data Privacy refers to the protection of personal information and how it is collected used, and shared. It ensures that individuals have control over their personal data and that organisations handle it responsibly.
- GDPR is a law that protects people's personal data and privacy.
- Country where it applies: European Union.

30. What is Data Ethics? Provide three examples of unethical use of data or AI, and explain what makes each example problematic.

- Data ethics refers to the responsible and fair use of data and AI.
- it ensures that data is used in a way that is honest, fair, and does not harm individuals or groups.

Example of unethical use: 1. Using personal data without consent
2. Biased hiring algorithms. 3. Spreading misinformation using AI

31. What is Data Governance? Define it and explain the key components of a good data governance framework.

- Data Governance is the ~~[illegible]~~ process of managing data availability, usability integrity and security within an organisation.
- Key components: Data quality management, Data security, policies and standards Data ownership.

32. What is Bias in AI? Explain what it means for an AI model to be biased. Provide a real-world example of AI bias and describe the potential consequences.

Bias in AI occurs when a model produces unfair or inaccurate results due to biased data or design.

Example: facial recognition system that performs poorly on certain skin tones.

Consequences:- Discrimination, unfair decisions; loss of trust.

33. What are the key differences between a Data Analyst, a Data Scientist, a Data Engineer, and an AI Engineer? Create a simple table comparing their roles, skills, and tools.

Role	Description	Skills	Tools
Data Analyst	Analyses data to find insight.	SQL, Excel, visualisation	Power BI, Excel
Data Scientist	Builds predictive models	Python, ML, statistic	Python, Scikit-learn
Data Engineer	Builds data systems	ETL, databases, cloud	AWS, Spark
AI Engineer:	Develops AI Systems	ML, Deep learning	TensorFlow, PyTorch

34. What is Business Intelligence (BI)? Name two popular BI tools (e.g., Power BI, Tableau) and explain how they help businesses make data-driven decisions.

* BI refers to tools and processes used to analyse data and support business decision-making.
* Power BI → creates dashboards and reports
- Tableau → data visualisation tool
* help businesses make informed decisions & turn data into insight.

Section E: Personal Reflection

Answer the following reflection questions thoughtfully and honestly. There is no right or wrong answer — we want to know what you think!

35. Which concept from this assignment did you find most interesting or surprising? Explain why.

The most interesting concept I found was AI, especially how machines can learn from data and make decisions. it is fascinating how AI is used in everyday tools like chatbots and recommendation systems.

36. Which concept did you find most difficult to understand? What resources did you use to help you understand it better?

The most difficult concept was understanding how neural network work. I used

Online resources such as Google's Machine Learning Crash Course and Youtube tutorials to better understand the topic.

37. How do you see Data and AI affecting your future career or everyday life? Be specific with examples.

Data and AI will have a big impact on my future career, especially as I am interested in Data Analytics. I will use these technologies to analyse data, build systems, and help businesses make better decisions

References

List all the sources you used to complete this assignment. Use the format below:

Author/Organisation. (Year). Title of article or webpage. Website Name. URL

1. IBM. (2023). What is a Database? https://www.ibm.com
2. AWS. (2023). What is a Data Warehouse? https://aws.amazon.com
3. Google Cloud. (2023). Data Analytics Basics https://cloud.google.com
4. Microsoft. (2023). Data Concepts https://learn.microsoft.com
5. Oracle. (2023). Data Mining Overview https://oracle.com
6. Open AI (2023). AI and LLM Overview https://openai.com
7. European Commission (2023). GDPR Overview https://ec.europa.eu
8. World Economic Forum. (2023). Data Ethics https://weforum.org

Good luck! We believe in you.

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